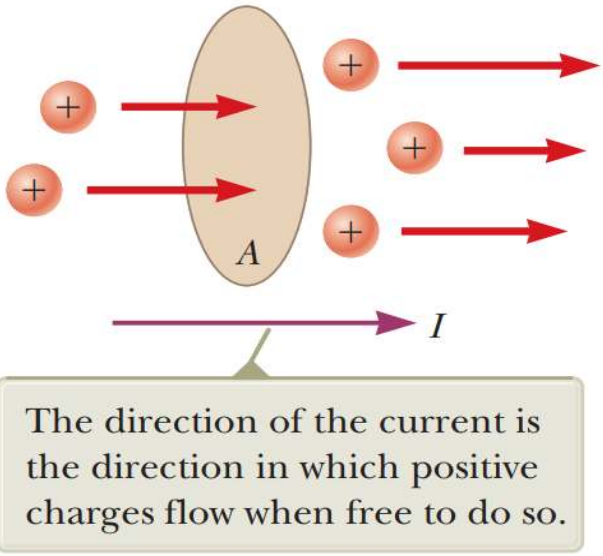
chapter (27):- Current and Resistance:-

27.1 Electric current:-



The direction of the current is the direction in which positive charges flow when free to do so.

flow of electric charge through a piece of material makes current.

Current:- the rate at which charge flows through asurface.

average current : the charge that passes through cross sectional area (A) Per unit time.

$$I_{avg} = \frac{\Delta Q}{\Delta t} \equiv [C/s] \equiv \text{ampere (A)}$$

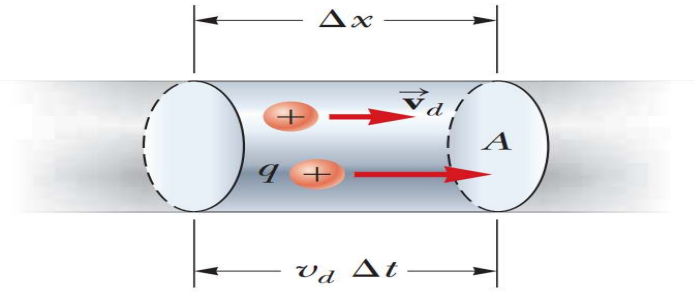
as $\Delta t \to 0$ we define instantanous current

$$I = \frac{dQ}{dt} \equiv [A]$$

the direction of the current is the same direction of +ve charges, and opposite the direction of electrons flow. Applying potential difference across wire creating electric field affects on the electrons

by electric force that moves electrons to make current.

Microscopic Model of Current:-



segment volume $V = A\Delta X$

Charge carrier:- moving charge (positive or negative).

Charge carriers density (n) = $\frac{\text{number of charge carriers}}{\text{Volume of the segment}}$

the total charge in the segment ΔQ

$$\Delta Q = (nA\Delta x)q$$

, q: is the charge on each carrier.

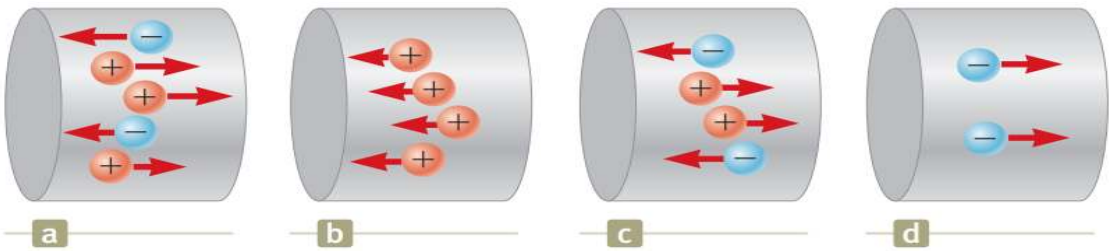
V_d :- velocity of the charge carriers. (drift speed)

$$V_d = \frac{\Delta x}{\Delta t} \Rightarrow \Delta x = V_d \Delta t$$

$$\frac{\Delta Q}{\Delta t} = \frac{(nA V_d \Delta t)q}{\Delta t}$$

$$I_{avg} = \frac{\Delta Q}{\Delta t} = nA V_d q$$

Quick quiz (27.1):- Rank the current from highest to lowest.



a, b and c, d

$a > b = c > d$.

Ex (27.1):- Drift speed in a copper wire:-

$A = 3.31 \times 10^{-6}\ m^2$ $\quad I = 10\ A$

each copper atom contributes one free electron

$\rho_{Cu} = 8.92\ g/cm^3$ $\quad$ molar mass of $Cu = 63.5\ g/mol$

One mole of substance = Avogadro's number of atoms (N_A)

$N_A = 6.02 \times 10^{23}\ mol^{-1}$

Volume of 1 mole of $Cu = \frac{\text{molar mass}}{\text{density}} = V$

electron density (n) in $Cu = \frac{N_A}{V} = \frac{N_A \rho}{M}$ $\quad \left| \rho = \frac{M}{V} \right.$ $\left. \therefore V = \frac{M}{\rho} \right|$

$I_{avg} = n q v_d A \Rightarrow v_d = \frac{I_{avg}}{n q A}$

$$V_d = \frac{IM}{N_A \rho q A} = \frac{10 \times 63.5 \times 10^{-3}}{6.02 \times 10^{23} \times 8.92 \times 10^{3} \times 1.6 \times 10^{-19} \times 3.31 \times 10^{-6}}$$

$$= 2.23 \times 10^{-4} \text{ m/s}$$

27.2 Resistance:-

$\vec{E} = 0$ inside a conductor only when it in electrostatic equalibrium.

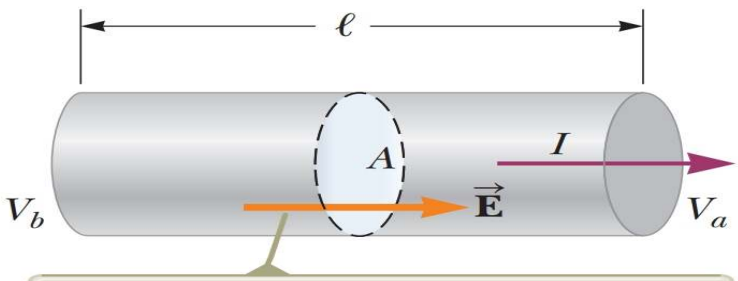
Current density ($\vec{J}$): is the current per unit area.

$$|\vec{J}| = \frac{I}{A} = \frac{nqv_d A}{A} = nqv_d \equiv [A/m^2]$$

$$\vec{J} \propto \vec{E}$$

$\vec{J} = \sigma \vec{E}$, where σ is the conductivity of the conductor.

general formula of Ohm's Law, material that obays this formula are called Ohmic materials and those don't obay that formula are nonohmic materials.



A potential difference $\Delta V = V_b - V_a$ maintained across the conductor sets up an electric field $\vec{E}$, and this field produces a current I that is proportional to the potential difference.

$$\Delta V = V_B - V_A$$

$$\Delta V = E L$$

$$J = \sigma E$$

$$J = \frac{\sigma \Delta V}{L} \Rightarrow \Delta V = \frac{L J}{\sigma}$$ ($J = \frac{I}{A}$)

$$\Delta V = \frac{L}{\sigma}\frac{I}{A} = \frac{L}{\sigma A} I$$

$$\Delta V = R I \quad , R = \frac{L}{\sigma A}$$

R :- resistance of conductor

$$R = \frac{\Delta V}{I} \equiv [\text{volt}/A] \equiv \text{ohm}(\Omega)$$

$$R = \frac{L}{\sigma A} \Rightarrow \sigma = \frac{L}{RA} \equiv \left[\frac{m}{\Omega \cdot m^2}\right] = (\Omega \cdot m)^{-1}$$

σ: Conductivity

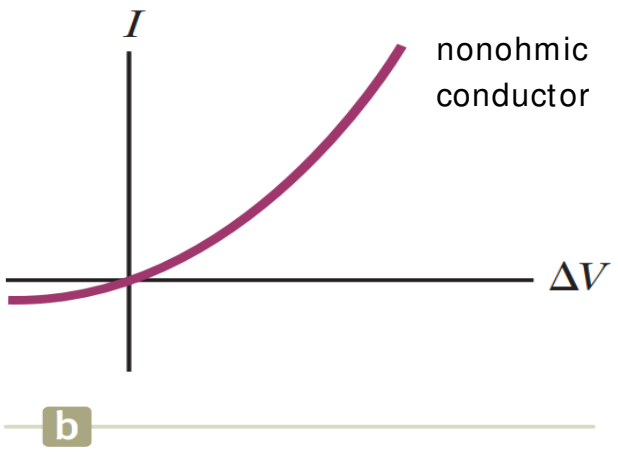
the inverse of conductivity is resistivity (ρ)

$$\rho = \frac{1}{\sigma} \equiv (\Omega \cdot m)$$

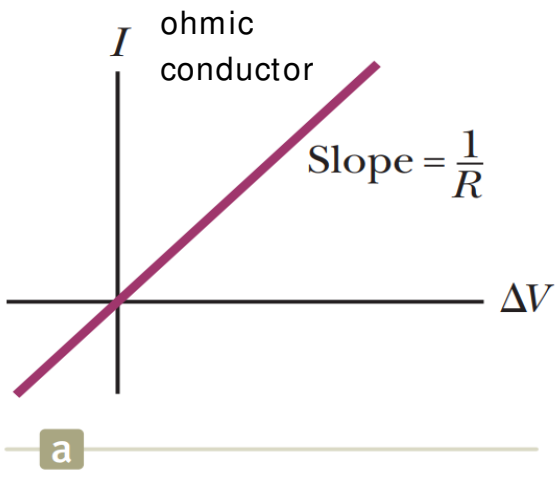
$$R = \frac{L}{\sigma A} = \frac{\rho L}{A}$$

ρ depends on the properties of material and temperature.

$R = \rho \frac{L}{A} \Rightarrow R \propto L \quad , R \propto \frac{1}{A}$



b



a

the relationship between ΔV and I is Linear when conductor is ohmic and nonlinear for nonohmic conductors

Quiz(27.2):- aconductor has radius r and length L, if both of them are doubled then the resistance:-

a- increase ⓑ- deerease c- remain the same.

Quiz(27.3):- for nonohmic conductor if we increase ΔV then R:-

a- increase ⓑ- decrease c- remain the same.

$r = 0.32\ mm$ $\rho = 1 \times 10^{-6}\ \Omega \cdot m$

a- $R = \frac{\rho L}{A}$

Resistance per unit length $\frac{R}{L} = \frac{\rho}{A}$

$A = \pi r^2$

$\frac{R}{L} = \frac{\rho}{\pi r^2} = \frac{1 \times 10^{-6}}{\pi (0.32 \times 10^{-3})^2} = 3.1\ \Omega/m$

b- $\Delta V = 10$ volt $L = 1m$

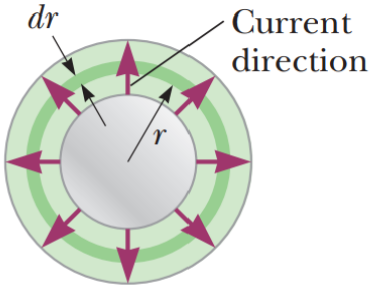
$\frac{R}{L} = 3.1 \Rightarrow R = L \times 3.1 = 1 \times 3.1 = 3.1\ \Omega$

$\Delta V = IR \Rightarrow I = \frac{\Delta V}{R} = \frac{10}{3.1} = 3.2\ A$

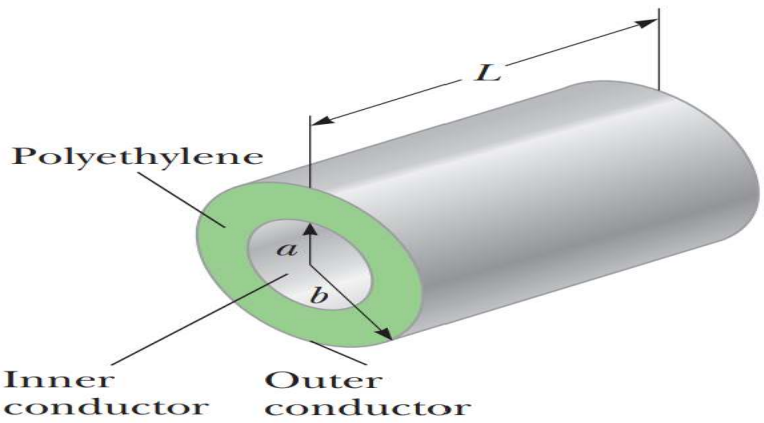
EX(27.3):- The radial Resistance of a Coaxial cable:-

$a = 0.5\ cm$ $L = 15\ cm$ $\rho = 1 \times 10^{13}\ \Omega \cdot m$

$b = 1.75\ cm$



b



a

the resistance of plastic shell $dR = \frac{\rho\, dr}{A}$

$$dR = \frac{\rho\, dr}{2\pi r L}$$

$$R = \int dR = \frac{\rho}{2\pi L} \int_a^b \frac{dr}{r} = \frac{\rho}{2\pi L} \ln\left(\frac{b}{a}\right)$$

$$R = \frac{1 \times 10^{13}}{(2\pi \times 0.15)} \ln\left(\frac{1.75}{0.5}\right) = 1.33 \times 10^{13}\ \Omega$$

$$R_{Cu} = \frac{\rho L}{A} = \frac{1.7 \times 10^{-8} \times 0.15}{\pi (0.5 \times 10^{-2})^2} = 3.2 \times 10^{-5}\ \Omega$$

$R_{Polyethylene} >>> R_{Cu}$; to prevent current leakage.

27.3 A Model for Electrical Conduction:-

electrical conduction has the following properties:

1- array of atoms plus free electrons (conduction electrons) make system of electrical conduction. where conducton electrons bound to their atoms.

2- in the absence of electric field, the conduction electrons move in random directions throug conductor

3- when the electric field is applied, the free electrons drift slowly in the direction that opposite to $\vec{E}$. $v_d << v_{avg}$

4- the electronsmotion after collision is independent of it's motion before collision. the work done on e^-'s by $\vec{E}$ stored as energy that transfered to atoms due to collisions, which increase the internal energy and temperature of the system.

for electron in an electric field, it will experience a force

$$\vec{F} = q\vec{E}$$

$$\vec{a} = \frac{\sum \vec{F}}{m_e} = \frac{q\vec{E}}{m_e} = \text{Constant}$$

$$\vec{v}_f = \vec{v}_i + \vec{a}t = \vec{v}_i + \frac{q\vec{E}}{m_e}t$$

$$\vec{v}_{f\,avg} = \vec{v}_d = 0 + \frac{q\vec{E}}{m_e}\tau$$

Where τ is the average time interval between successive collisions. τ depends on n and the atom size.

$$\vec{v}_d = \frac{q\vec{E}}{m_e}\tau$$

$$I_{avg} = nqv_dA$$

$$= nq\left(\frac{qE\tau}{m_e}\right)A$$

$$= \frac{nq^2E}{m_e}\tau A$$

$$J = \frac{I}{A} = \frac{nq^2E}{m_e}\tau \quad \text{and} \quad J = \sigma E$$

$$\Rightarrow \sigma = \frac{nq^2\tau}{m_e} \qquad \rho = \frac{1}{\sigma} = \frac{m_e}{nq^2\tau}$$

$\therefore$ σ and ρ don't depend on the strength of $\vec{E}$.

v_{avg} :- average speed between collisions.

mean free path (ℓ_{avg}) :- average distance between collisions.

$$\tau = \frac{\ell_{avg}}{v_{avg}}$$

27.4 Resistance and Temperature

the resistivity varies with temperature according to :-

$$\rho = \rho_0 [1 + \alpha (T - T_0)]$$

ρ :- resistivity at some temperature T.

ρ_0 :- " " " reference temperature

($T_0 = 20^\circ C$)

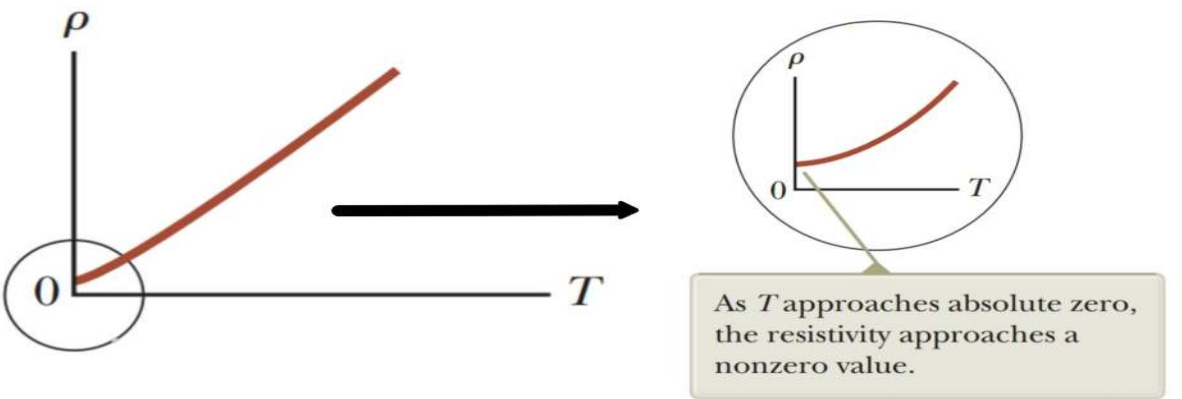
α :- temperature coefficient of resistivity.

$$\Delta\rho = \rho - \rho_0 \qquad \Delta T = T - T_0$$

$$\alpha = \frac{1}{\rho_0} \frac{\Delta\rho}{\Delta T} \equiv {}^\circ C^{-1}$$

R is proportional to ρ

$R = R_0[1 + \alpha(T - T_0)]$



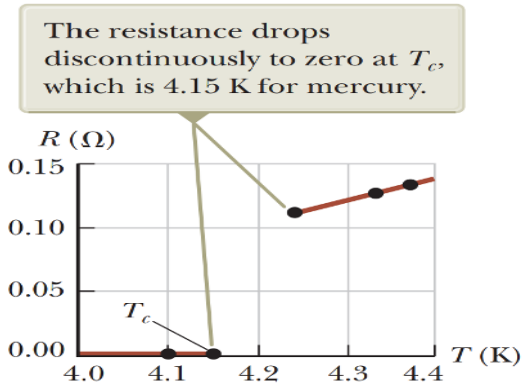
- at low temperature the relationship between ρ and T is nonlinear due to collisions of electrons with impurities and imperfections in the metal.
- at higher temperature the relationship between ρ and T is linear where the collisions are between electrons and metal atoms.

27.5 Superconductors:-

Superconductors are:- metals whose resistance decreases to zero when they are below a certain temperature T_c (critical temperature).

for mercury (Hg) $T_c = 4.15\,^{\circ}K$

below that temperature $R=0$.



because $R = 0$; current is persists without any applied ΔV. One of the important application of superconductors is in the development of super conducting magnets which is used in medical magnetic resonance imaging (MRI).

Ex:- Resistance dependence thermometer made of Platinium, it's resistance was 50 Ω at 20°C. if you put it in melt indium the resistance increases to (76.8) Ω, determine indium melting temperature? where $\alpha = 3.92 \times 10^{-3}\ ^{\circ}C^{-1}$

$$R = R_0[1 + \alpha(T - T_0)]$$

$$R = R_0 + R_0\alpha(T - T_0)$$

$$R - R_0 = R_0\alpha(T - T_0)$$

$$T = \frac{R - R_0}{R_0\alpha} + T_0$$

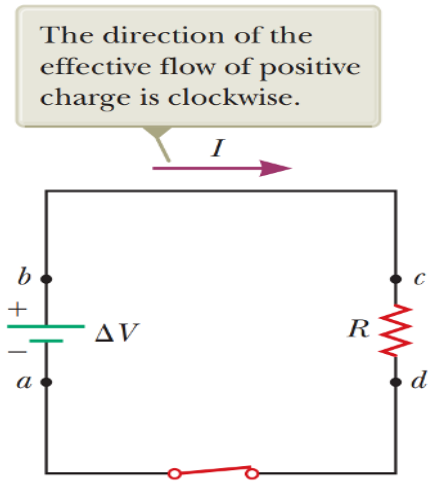
$$= \frac{76.8 - 50}{3.92 \times 10^{-3} \times 50} + 20$$

$$= 137 + 20$$

$$= 157\,^{\circ}C$$

27.6 Electrical Power:-

for charge @ moving from a → b from low potential to high potential ⇒ electric potential increases by $U = @\,\Delta V$ and chemical energy in the battery decreases by $@\,\Delta V$.



as the charge moves from c to d ⇒ U decreases because it's transfered to the resistor as heat.

the rate of loses electrical potential energy is Power (P), where

$$P = \frac{dU}{dt} = \frac{d}{dt}(@\,\Delta V) = I\,\Delta V \equiv [\text{watt (w)}]$$

$$P = I\,\Delta V = I^2 R = \frac{(\Delta V)^2}{R}$$

the process by which power is lost as internal energy in a conductor of resistance R is called Joule heating.

Ex(27.4):- Power in an Electric heater:-

$\Delta V = 120$ volt $\qquad R = 8\ \Omega$

a- $I = \frac{\Delta V}{R} = \frac{120}{8} = 15\ A$

b- $P = I^2 R$

$= (15)^2 (8)$

$= 1.8 \times 10^3$ watt

$= 1.8$ Kwatt

Ex(27.5):- Linking electricity and thermodynamics:-

$m_{water} = 1.5\ kg \qquad T_1 = 10^\circ C \qquad T_2 = 50^\circ C$

$t = 10$ minutes $\qquad \Delta V = 110$ volt $\qquad c = 4186\ J/kg.^\circ C$

a- Power $= \frac{\text{thermal energy } (Q)}{\text{time}} = \frac{Q}{\Delta t} = \frac{(\Delta V)^2}{R}$

$Q = mc\Delta T$, C is the specific heat for water

$Q = 1.5 \times 4186 \times (50 - 10) = 251160\ J$

$\frac{(\Delta V)^2}{R} = \frac{Q}{\Delta t} \Rightarrow R = \frac{(\Delta V)^2 \Delta t}{Q}$

$R = \frac{(110)^2 \times 10 \times 60}{251160} = 28.9\ \Omega$

b- Energy transferred (T) = $P \Delta t$

$$T = \frac{(\Delta V)^2}{R} \Delta t$$

$$= \frac{(110)^2}{28.9} \times \frac{10}{60} = 69.8 \text{ W·hr}$$

$$= 0.0698 \text{ Kw·hr}$$

Cost = T · Price of Kw·hr

$$= 0.0698 \times 0.11$$

$$= 0.008 \$$$